

1.1.47

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Problem

A feedback control system with high gain K is shown.

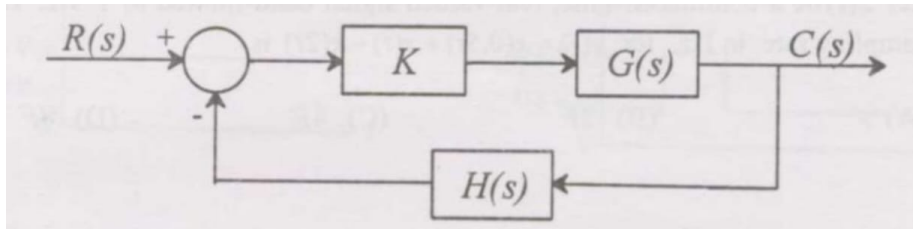


Figure:

The closed loop transfer function is :

- ① sensitive to perturbations in $G(s)$ and $H(s)$
- ② sensitive to perturbations in $G(s)$ but not to perturbations in $H(s)$
- ③ sensitive to perturbations in $H(s)$ but not to perturbations in $G(s)$
- ④ insensitive to perturbations in $G(s)$ and $H(s)$

System Description

$R(s) \Rightarrow$ Input

$C(s) \Rightarrow$ Output

$T(s) \Rightarrow$ closed loop transfer function

The system contain the following gains:

- Forward path: $KG(s)$
- Negative Feedback path: $H(s)$

Solution

At the summing junction

$$E(s) = R(s) - [H(s)]C(s) \quad (1)$$

So, now $E(s)$ goes through the system

$$C(s) = [KG(s)]E(s) \quad (2)$$

$$= KG(s) [R(s) - H(s)C(s)] \quad (3)$$

$$C(s) = KG(s)R(s) - KG(s)H(s)C(s) \quad (4)$$

$$C(s) + [KG(s)]H(s)C(s) = KG(s)R(s) \quad (5)$$

$$C(s) [1 + KG(s)H(s)] = KG(s)R(s) \quad (6)$$

Transfer Function

$$T(s) = \frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)} \quad (7)$$

$$T(s) = \frac{G(s)}{\frac{1}{K} + G(s)H(s)} \quad (8)$$

As K is very large (9)

$$T(s) \approx \frac{G(s)}{G(s)H(s)} \quad (10)$$

$$T(s) \approx \frac{1}{H(s)} \quad (11)$$

Sensitivity Analysis

- $G(s)$ cancels out $\Rightarrow$ insensitive to $G(s)$
- Output depends on $H(s)$
- Sensitive to perturbations in $H(s)$

Sensitive to perturbations in $H(s)$ but not in $G(s)$ (12)

Python Code 1

```
import numpy as np
import matplotlib.pyplot as plt

w = np.linspace(0.1, 10, 500)
s = 1j * w

K = 1e5 # very large gain

# Fixed G(s)
G = 1 / (s + 1)

# Strongly different H(s)
H1 = 1 / (s + 2)
H2 = 5 / (s + 0.2)

# Closed-loop transfer functions
T1 = (K * G) / (1 + K * G * H1)
T2 = (K * G) / (1 + K * G * H2)
```

Python Code 1

```
# Plot H(s)
plt.figure()
plt.plot(w, abs(H1), label=H1 = 1/(s+2))
plt.plot(w, abs(H2), '--', label=H2 = 5/(s+0.2))
plt.title(Large Difference in H(s))
plt.legend()
plt.grid()
plt.show()

# Plot T(s)
plt.figure()
plt.plot(w, abs(T1), label=T with H1)
plt.plot(w, abs(T2), '--', label=T with H2)
plt.title(Closed-loop T(s) (Clearly Different!))
plt.legend()
plt.grid()
plt.show()
```

Python Code 2

```
import numpy as np
import matplotlib.pyplot as plt

w = np.linspace(0.1, 10, 500)
s = 1j * w

K = 1e5 # large gain

# Fixed H(s)
H = 1 / (s + 2)

# Strongly different G(s)
G1 = 1 / (s + 1)
G2 = 5 / (s + 0.2)

# Closed-loop
T1 = (K * G1) / (1 + K * G1 * H)
T2 = (K * G2) / (1 + K * G2 * H)
```

Python Code 2

```
# Plot G(s)
plt.figure()
plt.plot(w, abs(G1), label=G1 = 1/(s+1))
plt.plot(w, abs(G2), '--', label=G2 = 5/(s+0.2))
plt.title(Large Difference in G(s))
plt.legend()
plt.grid()
plt.show()

# Plot T(s)
plt.figure()
plt.plot(w, abs(T1), label=T with G1)
plt.plot(w, abs(T2), '--', label=T with G2)
plt.title(Closed-loop T(s) (Still Almost Same!))
plt.legend()
plt.grid()
plt.show()
```

Plot Information

$$s = j\omega$$

For $H(s)$ changing :

$$G(s) = \frac{1}{s+1} \quad (13)$$

$$H_1(s) = \frac{1}{s+2} \quad (14)$$

$$H_2(s) = \frac{5}{s+0.2} \quad (15)$$

For $G(s)$ changing :

$$H(s) = \frac{1}{s+2} \quad (16)$$

$$G_1(s) = \frac{1}{s+1} \quad (17)$$

$$G_2(s) = \frac{5}{s+0.2} \quad (18)$$

Plot with $H(s)$ Changing

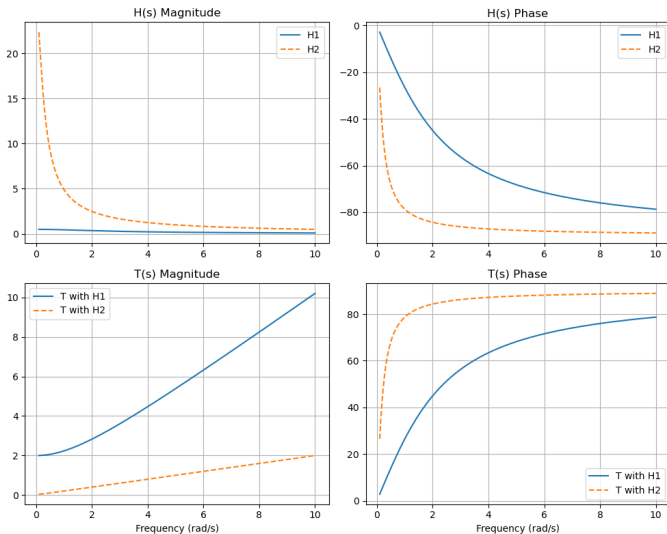


Figure: $H(s)$ is changing

Plot with $G(s)$ changing

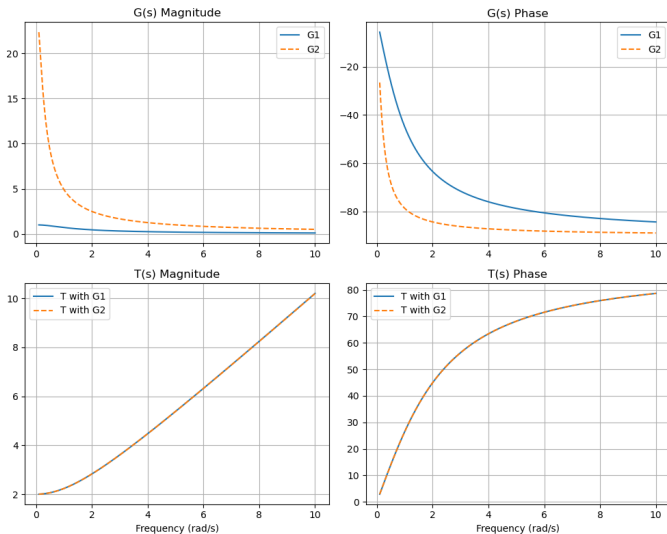


Figure: $G(s)$ Changing